

Vector Differentiation: Directional Derivative & Surface Angles

1. Directional Derivative

Name of Concept

Directional Derivative (D.D) and Maximum Directional Derivative.

Core Mathematical Formulas

The Directional Derivative of a scalar function ϕ in the direction of a vector $\vec{a}$ is given by:

$$\text{Directional Derivative} = \frac{\nabla\phi \cdot \vec{a}}{|\vec{a}|}$$

The Maximum Directional Derivative occurs in the direction of the gradient vector itself, and its magnitude is given by:

$$\text{Maximum Directional Derivative} = |\nabla\phi|$$

Beginner-Friendly Explanation of Operations

Vector differentiation allows us to calculate how functions change as we move through space.

- The Gradient ($\nabla\phi$):** The symbol ∇ (pronounced "del" or "nabla") represents the vector differential operator. When applied to a scalar function $\phi(x, y, z)$, it produces a vector called the gradient. It represents the rate and direction of the fastest spatial increase of the function.
 - The Unit Vector ($\frac{\vec{a}}{|\vec{a}|}$):** To check the rate of change in a *specific* direction specified by vector $\vec{a}$, we must eliminate the effect of the vector's length. We do this by dividing $\vec{a}$ by its absolute magnitude $|\vec{a}|$. This creates a unit vector (a vector with a length of 1) pointing in that exact direction.
 - The Dot Product ($\cdot$):** We compute the dot product between the gradient vector $\nabla\phi$ and the direction's unit vector. The result is a simple scalar number representing exactly how fast the function changes per unit distance in that direction.
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2. Angle Between Two Surfaces

Name of Concept

Angle of Intersection Between Two Scalar Surfaces.

Core Mathematical Formula

The angle θ between two intersecting surfaces at a specific point is equal to the angle between their normal vectors (gradients) at that point:

$$\cos\theta = \frac{\nabla u \cdot \nabla v}{|\nabla u||\nabla v|}$$

Beginner-Friendly Explanation of Operations

- Represent the Surfaces:** We rewrite the given surface equations in the form $u(x, y, z) = 0$ and $v(x, y, z) = 0$.
 - Find the Normal Vectors:** The gradient vector ∇u computed at a specific point is always mathematically perpendicular (normal) to the surface u at that point. By computing ∇u and ∇v , we obtain two geometric arrows sticking straight out of the two surfaces at their intersection point.
 - Calculate the Angle:** We use the geometric definition of the vector dot product to extract the cosine of the angle ($\cos\theta$) between these two normal arrows.
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3. Step-by-Step Problem Solution

Question

Find the angle between the surfaces $x \log z + 1 - y^2 = 0$ and $x^2 y + z = 2$ at the point $(1, 1, 1)$.

Step 1: Define the first scalar surface function

Let the first surface be defined as u :

$$u = x \log z + 1 - y^2$$

Step 2: Set up the gradient formula for the first surface

The gradient operator uses partial derivatives along the three coordinate axes:

$$\nabla u = \vec{i} \frac{\partial u}{\partial x} + \vec{j} \frac{\partial u}{\partial y} + \vec{k} \frac{\partial u}{\partial z}$$

Step 3: Compute the partial derivatives for u

When differentiating with respect to one variable, treat all other variables as constant numbers.

Differentiating u with respect to x treating y and z as constants:

$$\frac{\partial u}{\partial x} = \log z$$

Differentiating u with respect to y treating x and z as constants:

$$\frac{\partial u}{\partial y} = -2y$$

Differentiating u with respect to z treating x and y as constants:

$$\frac{\partial u}{\partial z} = x \cdot \frac{1}{z}$$

Step 4: Assemble the gradient vector field for u

Substitute the partial derivative expressions back into the gradient setup:

$$\nabla u = \vec{i}(\log z) + \vec{j}(-2y) + \vec{k} \left(\frac{x}{z} \right)$$

Step 5: Evaluate the gradient vector at the point $(1, 1, 1)$

Substitute $x = 1$, $y = 1$, and $z = 1$ into the vector expression. Note that $\log 1 = 0$:

$$\nabla u_{(1,1,1)} = \vec{i}(\log 1) + \vec{j}(-2(1)) + \vec{k} \left(\frac{1}{1} \right)$$

$$\nabla u_{(1,1,1)} = \vec{i}(0) - 2\vec{j} + \vec{k}$$

$$\vec{a} = \nabla u_{(1,1,1)} = -2\vec{j} + \vec{k}$$

This vector $\vec{a}$ represents the normal vector to the first surface at the given point.